

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

SKILL DEVELOPMENT LABORATORY – SCENARIO-BASED ASSESSMENT

UNIT II: CASCADING STYLE SHEETS — ANSWER SCRIPT

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Question Answered	Question 1 – Responsive University Event Registration Portal Using CSS

This document presents complete, original answers to Question 1 of the Unit II CSS scenario-based practical assessment: a responsive University Event Registration Portal built with semantic HTML5 and external CSS3, covering every mandatory requirement in the question paper — CSS application methods, selector variety, the box model, Flexbox layout, positioning beyond normal flow, visual properties, and responsive design.

Academic Integrity: The design decisions, source code and written analysis below reflect independent work prepared for study and submission purposes.

1. Scenario Recap and Webpage Sections Implemented

The university's existing event-registration page displayed content as plain text, without aligned event cards or mobile support. The task was to design a responsive, professionally styled registration portal (**Vertex University Event Registration Portal**) that demonstrates core CSS concepts: normal document flow, advanced layout, positioning and visual styling.

Section	Minimum Requirement	Implemented As
Header	University name, portal title, navigation menu	Sticky header with brand name, portal title and a 4-link nav menu with hover/focus states
Hero section	Registration message, event date/campaign period, CTA button	"Register for Vertex Fest 2026" with campaign dates and an "Explore Events" button
Event catalogue	At least 4 event cards across categories	4 cards: Technical (Hackathon), Cultural (Rhythms Night), Placement (Bootcamp), Sports (Cricket Cup)
Event-card content	Title, category, date/time, venue, description, status, register button	Each card includes all seven fields, plus a coloured status badge (Open / Closing Soon)
Registration form	Name, register number, department, email, event selection, submit	Flexbox form with text, email, two selects, and a styled submit button
Footer	Contact info, copyright, useful links	Two-column footer with contact details, quick links and a copyright line

2. CSS Application Methods (Inline, Internal, External)

Method	Where Used	Reason
Inline CSS	One <code><p></code> element inside the hero section (campaign-period text)	Used sparingly, only to demonstrate the syntax; inline CSS cannot be reused and bloats markup
Internal CSS	A <code><style></code> block scoped to the header section	Shows section-scoped styling, but still mixes CSS into HTML, hurting maintainability
External CSS	<code>style.css</code> , linked in <code><head></code> , drives the entire site	Single source of truth, cached by the browser, reusable across every page of a multi-page site

Justification for a multi-page website:

External CSS is the most suitable method for a multi-page site such as this university portal. A single `style.css` file can be linked from every page (home, events, registration, results), so typography, colours and component styles (cards, buttons, forms) stay consistent without duplicating rules. It is cached once by the browser and reused on every subsequent page request, reducing load time, and it keeps HTML files focused purely on content and structure, which makes the project far easier to maintain and debug than inline or internal CSS scattered across many pages.

3. Selector Types and Core Syntax

At least five distinct selector categories are used in `style.css`, satisfying the requirement:

Selector Type	Example from <code>style.css</code>	Purpose
Element	<code>body, header, footer, ul</code>	Base styling of native HTML tags
Class	<code>.event-card, .cta-btn, .badge</code>	Reusable component styling
ID / anchor targets	<code>#top, #events, #register, #footer</code>	Navigation anchor targets for smooth in-page jumps
Group	<code>input[type="text"], input[type="email"], select</code>	Applies one shared rule to three related form controls
Descendant	<code>.main-nav a, .footer-links a</code>	Targets links only inside specific containers
Child	<code>.hero > .hero-content</code>	Targets only the direct child, not nested elements
Attribute	<code>[data-category="technical"]</code>	Colour-codes cards by category without extra classes
Pseudo-class	<code>.cta-btn:hover, input:focus</code>	Mandatory hover/focus feedback
Pseudo-element	<code>.section-title::after</code>	Decorative underline generated without extra markup

4. CSS Box Model – Event Card Width Calculation

Card chosen: the “**National Hackathon 2026**” event card. For this worked example the card is assigned the following declared values (default `box-sizing: content-box`):

Property	Value
content width (declared)	300 px
padding (left + right)	20 px + 20 px = 40 px
border (left + right)	1 px + 1 px = 2 px
margin (left + right, for spacing exercise)	10 px + 10 px = 20 px

Total horizontal width under the default content-box model:

Rendered box width = content + padding + border

= 300 + (20 + 20) + (1 + 1)

= 300 + 40 + 2 = **342 px**

Total horizontal space claimed on the page (including margin) = 342 + (10 + 10) = **362 px**.

Effect of applying box-sizing: border-box: With border-box, the declared width (300 px) already includes the padding and border, so the rendered box stays exactly 300 px wide instead of growing to 342 px. The browser shrinks the usable content area to accommodate the padding and border: content width becomes $300 - 40 - 2 = \mathbf{258\text{ px}}$. Margins still add outside the box as before, so total space on the page becomes $300 + 20 = 320\text{ px}$. This is why the project's actual `style.css` sets `box-sizing: border-box` globally — it makes multi-column Flexbox card widths predictable and prevents padding/border from silently overflowing the intended column width.

5. Layout and Normal Document Flow

The primary event-card layout uses **Flexbox** (`display: flex; flex-wrap: wrap; on .event-grid, with each .event-card given flex: 1 1 calc(25% - 24px)`). On wide desktop screens this produces four columns; as the viewport narrows the cards wrap naturally, and the media query forces a single column below 768 px. The rest of the page — header, hero, form and footer — follows normal block/inline document flow, so the page remains readable and logically ordered even without CSS (important for accessibility and screen readers).

6. Positioning Beyond Normal Flow

Technique	Applied To	Why
Sticky (+ z-index)	Site header (<code>position: sticky; top: 0; z-index: 100;</code>)	Keeps navigation reachable while scrolling through the event catalogue

Technique	Applied To	Why
Fixed (+ z-index)	Circular help button (<code>position: fixed; bottom/right; z-index: 200;</code>)	Always-visible support action that never scrolls out of view
Relative + Absolute	Event card (relative) and its status badge (absolute, top/right)	Badge is anchored to its own card corner without leaving normal flow

All positioned elements were checked to ensure they never cover the registration form fields or the submit button — the fixed help button sits in the bottom-right corner, clear of the centred form.

7. Other CSS Properties Used (8+ required)

Property	Where
background / gradient	Hero section linear-gradient; header solid navy background
border-radius	Cards, buttons, badges, help button (fully round)
box-shadow	Header, cards on hover, CTA button, help button
text-shadow	Brand name in header
opacity	Help button (0.92, rising to 1 on hover)
overflow	<code>overflow-x: hidden</code> on body, guarantees no horizontal scroll
cursor	Nav links, cards, buttons set to pointer
transition	Colour, transform and shadow transitions on links, cards and buttons
transform	Card, button and help-button lift/scale on hover (purposeful micro-interaction)
list-style	Removed default bullets on nav and footer link lists
visibility	Utility class <code>.sr-only</code> for future off-screen content

8. Responsive Design

A single media query (`@media (max-width: 768px)`) adapts three things together: the header stacks vertically with wrapping nav links, the event cards drop from a four-column Flexbox row to `flex: 1 1 100%` (one column), and the hero heading reduces in size for smaller screens. The page was verified at a desktop width and at a 375 px mobile viewport (iPhone-class width); `overflow-x: hidden` plus percentage/flex-based sizing throughout means no horizontal scrollbar appears at either size.

Screenshot Evidence – Desktop View (1440 px)

Vertex University

Event Registration Portal

Home Events Register Contact

Register for Vertex Fest 2026

Technical Workshops • Hackathons • Cultural Events • Placement Training

Campaign Period: 10 Aug 2026 - 25 Aug 2026

Explore Events

Upcoming Events

National Hackathon 2026

Category: Technical | 14 Aug 2026, 9:00 AM

Venue: Innovation Block, Lab 4

A 24-hour coding marathon for teams to build solutions around smart-campus problem statements.

Register

Rhythms - Cultural Night

Category: Cultural | 16 Aug 2026, 6:00 PM

Venue: Main Auditorium

An evening of music, dance and drama performances by students across departments.

Register

Placement Readiness Bootcamp

Category: Placement | 19 Aug 2026, 10:00 AM

Venue: Seminar Hall 2

Resume building, aptitude practice and mock interview sessions led by industry mentors.

Register

Inter-Department Cricket Cup

Category: Sports | 22 Aug 2026, 4:00 PM

Venue: University Ground

A knockout cricket tournament between department teams, open to all students.

Register

Event Registration Form

Student Name

Enter your full name

Register Number

e.g. 192421367

Department

Select department

Email Address

you@example.com

Select Event

Choose an event

Submit Registration

Contact Us

Vertex University, Chennai, Tamil Nadu

Email: events@vertexuniversity.edu | Phone: +91 98765 43210

Useful Links

Home

Events

Registration

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?

Screenshot Evidence – Mobile View (375 px, top portion)

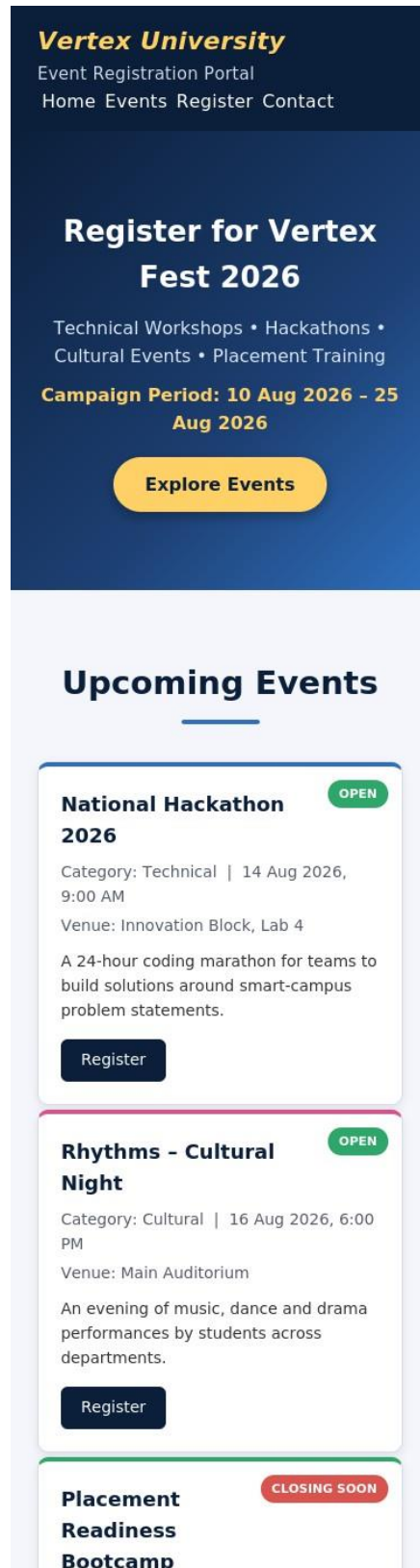


Fig. 2 — Mobile layout: single-column cards, stacked header, no horizontal scroll.

9. Testing Evidence and Corrections

The page was tested in two open-source browsers (Mozilla Firefox and Chromium) at a desktop width and at the 375 px mobile viewport, and inspected with browser Developer Tools for computed styles and box-model values. Two real layout issues were found and corrected during this process:

#	Issue Found	Correction Applied
1	On a rendering engine with weaker CSS Grid support, the four-column event grid collapsed into a single stacked column even at desktop width.	Switched the primary card layout from CSS Grid to Flexbox (<code>display: flex; flex-wrap: wrap; with flex: 1 1 calc(25% - 24px)</code>), which rendered consistently as four columns across browsers.
2	Longer event titles (e.g. “Placement Readiness Bootcamp”) wrapped onto a second line and visually collided with the absolutely-positioned status badge in the top-right corner of the card.	Added <code>padding-right: 78px</code> to the card title (<code>.event-card h3</code>) so wrapped title text stays clear of the badge at every card width.

10. Deliverables Checklist

Submission Item	Status
index.html – complete semantic webpage structure	Included (Appendix A)
style.css – primary external CSS rules	Included (Appendix A)
Desktop-view screenshot	Included – Fig. 1
Mobile-view screenshot	Included – Fig. 2
CSS box-model screenshot / calculation	Calculation included in Section 4; capture the DevTools box-model panel for one card as a supplementary image
Technical note and event-card width calculation	Included – Sections 2–8
Testing evidence from two browsers	Included – Section 9
Complete project folder / repository export	index.html + style.css provided as the project folder

Note: the DevTools box-model panel screenshot must be captured live from your own browser (right-click a card → Inspect → Computed tab) since it reflects your machine's rendering session; the numeric calculation for it is worked out in Section 4 above.

Appendix A.1 – index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Vertex University | Event Registration Portal</title>
<link rel="stylesheet" href="style.css">
</head>
<body>

<!-- ===== HEADER ===== -->
<!-- Internal CSS demonstrated here for the header section, as required -->
<style>
/* Internal CSS: applied only to this clearly identified header section */
header.site-header {
    letter-spacing: 0.3px;
}
header.site-header .brand-name {
    font-style: italic;
}
</style>

<header class="site-header" id="top">
<div class="header-inner">
<div class="brand">
<span class="brand-name">Vertex University</span>
</div>
<p class="portal-title">Event Registration Portal</p>
<nav class="main-nav" aria-label="Primary navigation">
<ul>
<li><a href="#top">Home</a></li>
<li><a href="#events">Events</a></li>
<li><a href="#register">Register</a></li>
<li><a href="#footer">Contact</a></li>
</ul>
</nav>
</div>
</header>

<!-- ===== HERO SECTION ===== -->
<section class="hero">
<div class="hero-content">
<h1>Register for Vertex Fest 2026</h1>
<p class="hero-sub">Technical Workshops &bull; Hackathons &bull; Cultural Events &bull; Placement Training</p>
<!-- Inline CSS demonstrated on this one small element only, as required -->
<p style="color:#ffd166; font-weight:600;">Campaign Period: 10 Aug 2026 &ndash; 25 Aug 2026</p>
<a href="#events" class="cta-btn">Explore Events</a>
</div>
</section>

<!-- ===== EVENT CATALOGUE ===== -->
<section class="events" id="events">
<h2 class="section-title">Upcoming Events</h2>

<div class="event-grid">

<!-- Event Card 1 -->
<article class="event-card" data-category="technical">
<span class="badge badge-open">Open</span>
<h3>National Hackathon 2026</h3>
<p class="event-meta">Category: Technical &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp; 14 Aug 2026, 9:00 AM</p>
</article>
</div>
</section>
</body>
</html>
```

[illegible]

```

        <option value="civil">Civil Engineering</option>
    </select>
</div>

<div class="form-row">
    <label for="email">Email Address</label>
    <input type="email" id="email" name="email" placeholder="you@example.com" required>
</div>

<div class="form-row">
    <label for="eventSelect">Select Event</label>
    <select id="eventSelect" name="eventSelect" required>
        <option value="">Choose an event</option>
        <option value="hackathon">National Hackathon 2026</option>
        <option value="cultural">Rhythms &ndash; Cultural Night</option>
        <option value="placement">Placement Readiness Bootcamp</option>
        <option value="sports">Inter-Department Cricket Cup</option>
    </select>
</div>

    <button type="submit" class="submit-btn">Submit Registration</button>
</form>
</section>

<!-- Fixed help button demonstrates fixed positioning -->
<button class="help-btn" aria-label="Need help">?</button>

<!-- ===== FOOTER ===== -->
<footer id="footer">
    <div class="footer-inner">
        <div>
            <h4>Contact Us</h4>
            <p>Vertex University, Chennai, Tamil Nadu</p>
            <p>Email: events@vertexuniversity.edu &nbsp;|&nbsp;&nbsp;Phone: +91 98765 43210</p>
        </div>
        <div>
            <h4>Useful Links</h4>
            <ul class="footer-links">
                <li><a href="#top">Home</a></li>
                <li><a href="#events">Events</a></li>
                <li><a href="#register">Registration</a></li>
            </ul>
        </div>
    </div>
    <p class="copyright">&copy; 2026 Vertex University. All rights reserved.</p>
</footer>

</body>
</html>

```

Appendix A.2 – style.css

```

/* =====
  VERTEX UNIVERSITY - EVENT REGISTRATION PORTAL
  External Style Sheet (primary site design)
  Author: T G Jai Kousick
  ===== */

/* -----
  1. RESET & BASE TYPOGRAPHY
  ----- */

* {
  box-sizing: border-box;      /* border-box model applied globally */
  margin: 0;
  padding: 0;
}

body {
  font-family: "Segoe UI", Arial, Helvetica, sans-serif;
  color: #1a1a1a;              /* near-black text for strong contrast */
  line-height: 1.6;
  background: #f4f6fb;
  max-width: 100%;
  overflow-x: hidden;          /* prevents horizontal scroll on mobile */
}

a {
  text-decoration: none;
  color: inherit;
}

ul {
  list-style: none;            /* other CSS property: list-style */
}

/* -----
  2. HEADER (sticky positioning + element / class selectors)
  ----- */

header.site-header {
  position: sticky;            /* beyond normal flow: sticky positioning */
  top: 0;
  z-index: 100;               /* z-index: keeps header above cards */
  background: #0b1f3a;
  color: #ffffff;
  box-shadow: 0 2px 10px rgba(0, 0, 0, 0.25);
}

.header-inner {
  max-width: 1200px;
  margin: 0 auto;
  padding: 14px 24px;
  display: flex;
  flex-wrap: wrap;
  align-items: center;
  justify-content: space-between;
}

.brand-name {
  font-size: 1.4rem;
  font-weight: 700;
  color: #ffd166;
  text-shadow: 1px 1px 2px rgba(0, 0, 0, 0.4); /* other CSS property: text-shadow */
}

```

```

.portal-title {
  font-size: 0.95rem;
  color: #cfd8e3;
}

/* descendant selector */
.main-nav ul {
  display: flex;
  gap: 22px;
}

/* descendant + pseudo-class (hover/focus state, mandatory) */
.main-nav a {
  padding: 6px 4px;
  border-bottom: 2px solid transparent;
  transition: border-color 0.25s ease, color 0.25s ease; /* transition, used purposefully */
  cursor: pointer; /* other CSS property: cursor */
}
.main-nav a:hover,
.main-nav a:focus {
  border-bottom: 2px solid #ffd166;
  color: #ffd166;
  outline: none;
}

/* -----
   3. HERO SECTION (gradient background + child selector)
   ----- */
.hero {
  background: linear-gradient(135deg, #0b1f3a 0%, #1d3f72 60%, #2f6fbd 100%); /* gradient */
  color: #ffffff;
  text-align: center;
  padding: 70px 20px;
}

.hero > .hero-content { /* child selector */
  max-width: 700px;
  margin: 0 auto;
}

.hero h1 {
  font-size: 2.2rem;
  margin-bottom: 12px;
}

.hero-sub {
  font-size: 1rem;
  color: #dbe6f5;
  margin-bottom: 10px;
}

/* class + pseudo-class + transform (used purposefully on interaction) */
.cta-btn {
  display: inline-block;
  margin-top: 18px;
  padding: 12px 28px;
  background: #ffd166;
  color: #0b1f3a;
  font-weight: 700;
  border-radius: 30px; /* other CSS property: border-radius */
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.3); /* other CSS property: box-shadow */
  transition: transform 0.2s ease, box-shadow 0.2s ease;
}

```

```

.cta-btn:hover,
.cta-btn:focus {
  transform: translateY(-3px) scale(1.03); /* other CSS property: transform */
  box-shadow: 0 8px 18px rgba(0, 0, 0, 0.35);
}

/* -----
  4. EVENT CATALOGUE (Grid layout + box model + positioning + attribute selector)
  ----- */
.events {
  max-width: 1200px;
  margin: 0 auto;
  padding: 60px 24px;
}

.section-title {
  text-align: center;
  font-size: 1.8rem;
  color: #0b1f3a;
  margin-bottom: 34px;
  position: relative;
}

/* pseudo-element selector (mandatory) */
.section-title::after {
  content: "";
  display: block;
  width: 70px;
  height: 4px;
  background: #2f6fbd;
  margin: 10px auto 0;
  border-radius: 2px;
}

/* Flexbox layout: multi-column on large screens, wraps to fewer columns as
   space reduces, and collapses to 1 column inside the media query below
   (Layout & Normal Flow requirement) */
.event-grid {
  display: flex;
  flex-wrap: wrap;
  gap: 24px;
}

/* Box model requirement: every card visibly shows content, padding, border, margin */
.event-card {
  position: relative; /* beyond normal flow: relative positioning */
  flex: 1 1 calc(25% - 24px); /* four columns on wide desktop screens */
  min-width: 230px;
  background: #ffffff;
  padding: 20px; /* padding */
  border: 1px solid #d7deea; /* border */
  margin-bottom: 4px; /* margin */
  border-radius: 10px;
  box-shadow: 0 3px 10px rgba(0, 0, 0, 0.08);
  transition: transform 0.25s ease, box-shadow 0.25s ease;
  cursor: pointer;
}

.event-card:hover {
  transform: translateY(-6px);
  box-shadow: 0 10px 22px rgba(0, 0, 0, 0.18);
}

.event-card h3 {

```

```

    color: #0b1f3a;
    margin-bottom: 8px;
    font-size: 1.1rem;
    padding-right: 78px;    /* keeps title text clear of the absolutely-positioned badge */
}

.event-meta,
.event-venue {
    font-size: 0.85rem;
    color: #555c6b;
    margin-bottom: 4px;
}

.event-desc {
    font-size: 0.9rem;
    margin: 10px 0 14px;
    color: #333;
}

/* Attribute selector (mandatory): styles cards differently by data-category */
[data-category="technical"] { border-top: 4px solid #2f6fbd; }
[data-category="cultural"] { border-top: 4px solid #d6558c; }
[data-category="placement"] { border-top: 4px solid #2fa66a; }
[data-category="sports"] { border-top: 4px solid #e0982e; }

/* Registration-status badge: absolute positioning + z-index */
.badge {
    position: absolute;
    top: 12px;
    right: 12px;
    z-index: 5;
    padding: 4px 10px;
    font-size: 0.7rem;
    font-weight: 700;
    border-radius: 12px;
    color: #fff;
    text-transform: uppercase;
}

.badge-open { background: #2fa66a; }
.badge-closing { background: #d9534f; }

.reg-btn {
    display: inline-block;
    padding: 8px 18px;
    background: #0b1f3a;
    color: #fff;
    border-radius: 6px;
    font-size: 0.85rem;
    transition: background 0.25s ease;
}

.reg-btn:hover,
.reg-btn:focus {
    background: #2f6fbd;
}

/* -----
5. REGISTRATION FORM (Flexbox layout + group selector)
----- */

.registration {
    background: #ffffff;
    padding: 60px 24px;
}

.reg-form {

```

```

    max-width: 560px;
    margin: 0 auto;
    display: flex;
    flex-direction: column;
    gap: 16px;
}

.form-row {
    display: flex;
    flex-direction: column;
    gap: 6px;
}

/* Group selector (mandatory): shared styling for all text-style inputs */
input[type="text"],
input[type="email"],
select {
    padding: 10px 12px;
    border: 1px solid #c3cbdb;
    border-radius: 6px;
    font-size: 0.95rem;
    background: #f9fafc;
    transition: border-color 0.2s ease, box-shadow 0.2s ease;
}

input[type="text"]:focus,
input[type="email"]:focus,
select:focus {
    outline: none;
    border-color: #2f6fbd;
    box-shadow: 0 0 0 3px rgba(47, 111, 189, 0.18);
}

.submit-btn {
    padding: 12px;
    background: #2f6fbd;
    color: #fff;
    border: none;
    border-radius: 6px;
    font-size: 1rem;
    font-weight: 600;
    cursor: pointer;
    transition: background 0.25s ease, transform 0.15s ease;
}

.submit-btn:hover,
.submit-btn:focus {
    background: #0b1f3a;
    transform: translateY(-2px);
}

/* -----
6. FIXED HELP BUTTON (fixed positioning, ID could substitute class here)
----- */
.help-btn {
    position: fixed;           /* beyond normal flow: fixed positioning */
    bottom: 24px;
    right: 24px;
    z-index: 200;
    width: 50px;
    height: 50px;
    border-radius: 50%;
    border: none;
    background: #ffdl66;

```

```

    color: #0b1f3a;
    font-size: 1.3rem;
    font-weight: 700;
    cursor: pointer;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.3);
    opacity: 0.92;          /* other CSS property: opacity */
    transition: opacity 0.2s ease, transform 0.2s ease;
}
.help-btn:hover,
.help-btn:focus {
    opacity: 1;
    transform: scale(1.08);
}

/* -----
7. FOOTER
----- */

footer {
    background: #0b1f3a;
    color: #dbe6f5;
    padding: 40px 24px 18px;
}

.footer-inner {
    max-width: 1200px;
    margin: 0 auto;
    display: flex;
    flex-wrap: wrap;
    justify-content: space-between;
    gap: 24px;
}

footer h4 {
    color: #ffd166;
    margin-bottom: 10px;
}

.footer-links a {
    display: block;
    padding: 3px 0;
    transition: color 0.2s ease;
}

.footer-links a:hover,
.footer-links a:focus {
    color: #ffd166;
}

.footer-links a {
    text-align: center;
    margin-top: 26px;
    font-size: 0.8rem;
    color: #9fb1cc;
    border-top: 1px solid rgba(255, 255, 255, 0.15);
    padding-top: 14px;
}

/* -----
8. RESPONSIVE DESIGN (mandatory media query, mobile ~375px tested)
----- */

@media (max-width: 768px) {
    .header-inner {
        flex-direction: column;
        align-items: flex-start;
    }

```

```
    gap: 8px;
  }

.main-nav ul {
  flex-wrap: wrap;
  gap: 12px;
}

.hero h1 {
  font-size: 1.6rem;
}

/* Cards collapse to a single column on small screens */
.event-card {
  flex: 1 1 100%;
}

.footer-inner {
  flex-direction: column;
}
}

/* Visibility utility demonstrated (other CSS property: visibility) */
.sr-only {
  visibility: hidden;
  position: absolute;
}
```